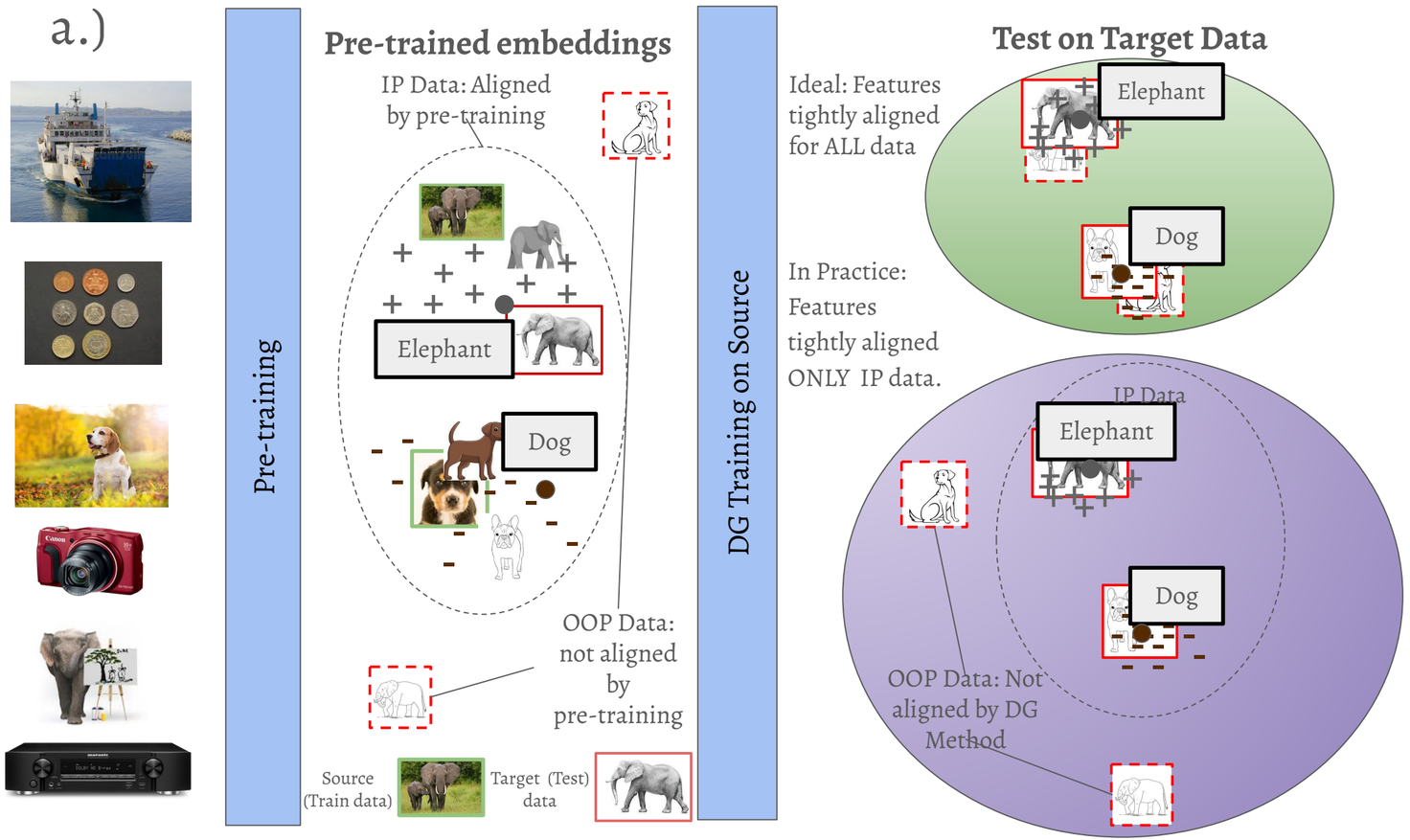
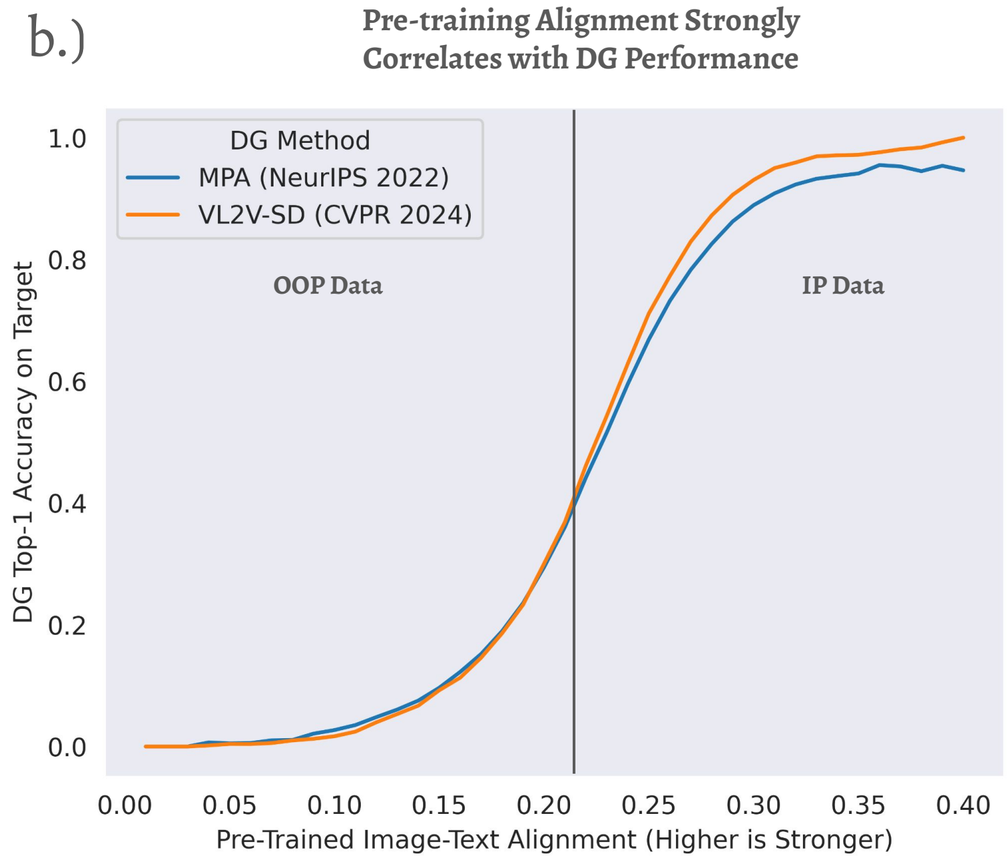

a.)
Pre-training
Pre-trained embeddings
IP Data: Aligned by pre-training
Elephant
Dog
OOP Data: not aligned by pre-training
Source (Train data)
Target (Test) data
DG Training on Source
Test on Target Data
Ideal: Features tightly aligned for ALL data
Elephant
Dog
In Practice: Features tightly aligned ONLY IP data.
IP Data
Elephant
Dog
OOP Data: Not aligned by DG Method


b.)
Pre-training Alignment Strongly Correlates with DG Performance
DG Method
MPA (NeurIPS 2022)
VL2V-SD (CVPR 2024)
OOP Data
IP Data
DG Top-1 Accuracy on Target
1.0
0.8
0.6
0.4
0.2
0.0
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
0.40
Pre-Trained Image-Text Alignment (Higher is Stronger)